ACE21 16S indicator analysis

Comparison of Hannah's historical output, a corrected rerun on Hannah's RData, and the current integrated dataset

Bottom line

The current integrated analysis is reproducible. Hannah's saved indicator table is not reproducible from her script as written because the script filters **Treatment == 'T0'**, while Treatment contains only Ambient and High. The script also documents that the statistical summary was copied manually into the CSV.

What was compared

To distinguish a coding/provenance problem from changes in sample membership, the indicator analysis was rerun three ways: (1) Hannah's manually saved CSV, (2) a corrected direct rerun on Hannah's original *16s_phyloseq4HE.RData* using **Time.Point == 'T0'**, and (3) the integrated analysis using the current main-script *Bac.seq* object.

Coral species	Historical saved CSV	Corrected historical rerun	Current integrated	Corrected/current shared
Montipora capitata	32	15	16	15
Pavona varians	52	12	8	5
Pocillopora acuta	71	37	33	33
Porites compressa	13	8	8	8

Interpretation of agreement

The corrected historical rerun and current analysis agree very closely for Montipora, Pocillopora, and Porites. The historical/current Jaccard similarities were 0.94, 0.89, and 1.00, respectively. This shows that most of the large apparent disagreement came from comparing a manually transcribed historical table with directly generated results.

Pavona is the exception. Its Jaccard similarity was 0.33. Two Pavona T0 samples, **S208** and **SN35b**, were deliberately excluded from the current dataset. The historical dataset contained only 14 Pavona T0 samples, so removing two observations can substantially alter permutation-based indicator results.

Relevant sample-set changes

Sample	Species	Time point	Current status	Indicator-analysis consequence
S1753b	Montipora	T0	Excluded	Changes the Montipora T0 group
S208	Pavona	T0	Excluded	Changes the Pavona T0 group
SN35b	Pavona	T0	Excluded	Changes the Pavona T0 group
S3476	Porites	TF	Excluded	No direct effect on the T0 test
S3059	Pocillopora	T1	Restored	No direct effect on the T0 test

Multiple-testing result

The current *multipatt()* result contains 2,437 rows and 2,432 non-missing permutation tests. At raw p <= 0.05, 79 associations were detected across all group patterns; 65 were associations with one individual coral species. After Benjamini-Hochberg correction, **zero associations remained significant**. The smallest raw p-value was 0.001 and the smallest adjusted p-value was approximately 0.270.

Recommended interpretation

Use the directly generated integrated output rather than the manually transcribed historical table. Add adjusted p-values to the exported results. Treat the 65 raw-p associations as exploratory candidates rather than statistically significant indicator taxa. If indicator analysis remains a project objective, choose a biologically justified prevalence or abundance filter before testing so extremely rare OTUs are removed and the number of tests is reduced. The filtering rule should be selected independently of which result appears most favorable.

Analysis files were not changed during this examination. This memo records the comparison performed on September 2, 2026.